

Overview

The layout generator now supports richer filtering. The `CodeCityFilters` interface was extended to replace the single `symbolKind` string with a `symbolKinds: SymbolKindGroup[]` array and to add a `changeStates: CodeCityChangeState[]` array. New helper functions `normalizeRelPath` and `fileMatchesSubsystem` were added to centralise path handling. A `districtScoped` flag (lines 171-175) short-circuits subsystem/extension checks when the filters are all "all".

Key changes

- **Imports** (lines 1-5): `changeStateMatchesFilters`, `symbolMatchesKindGroups`, and `SymbolKindGroup` from `./docsExploreFilters`.
- **Interface update** (lines 22-25): `symbolKinds` and `changeStates` added; `symbolKind` removed (line 17).
- **Path helpers** (lines 120-135): `normalizeRelPath` and `fileMatchesSubsystem`.
- **Visibility logic** (lines 178-181): `fileMatchesSubsystem` replaces direct path checks.
- **Symbol kind filtering** (lines 193-196): uses `symbolMatchesKindGroups`.
- **Change-state filtering** (lines 196-198): uses `changeStateMatchesFilters`.
- **Author & search** (lines 199-206, 210-214): author match logic and a case-insensitive search over file path, symbol name, and kind.
- **Result shape** (lines 317-322): returns `districts`, `buildings`, `subsystems`, `extensions`, and `symbolKinds`.

Impact

- **Correctness:** Filters now honor grouped symbol kinds and change states; subsystem matching handles both relative and absolute paths.
- **Maintainability:** Centralised path logic reduces duplication; the filter interface more clearly expresses intent.
- **Compatibility:** Code that still references `symbolKind` will fail to compile; UI and tests must migrate to `symbolKinds`.
- **Performance:** Minor overhead from helper calls; `districtScoped` cuts unnecessary checks for large models.

Risks & follow-ups

- **API breakage:** Verify all callers update from `symbolKind` to `symbolKinds`.
- **Subsystem logic:** Test `fileMatchesSubsystem` on nested subsystems and mixed path separators.
- **Empty filter arrays:** Ensure `symbolMatchesKindGroups` and `changeStateMatchesFilters` treat empty arrays as "all".

- **Regression:** Run existing layout unit tests to confirm visual and data integrity.